# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-LINSOLVE-EN · v2.1 · 2026-11-01 |
| Package | star-linsolve |
| Version | 0.1.0 |
| Capability | star.numerics.linsolve |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_linsolve |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_linsolve-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_linsolve-0.1.0-py3-none-any.whl | 100adaeb787c36a5c4f333dda8edf042140f20edd5b1f16dc23705f07bd56feb |
| star_linsolve.py | d48a71831dd3ba433b1eced9bdf0f1dc07557a69a3140515c53404bd6f93ead3 |
| star_linsolve-0.1.0-source.zip | 80d823829c9566e99f1f46ade0018244626a9f7d0d034d57823f6c4b8b62e79c |

# 5. What it does, and its limits

Determinant, solution and inverse of a square matrix of order 1 to 20 by elimination in exact rational arithmetic, each result rounded once: conditioning costs no digit and singularity is decided exactly; equal to SymPy's exact results on all 151 systems of the cross-check, Hilbert matrices of order 6 to 12 included.

Exact arithmetic on the floats you pass: 0.1 is not one tenth, and the solution of a badly conditioned system still changes a lot when the data change a little; the answer is exact for the matrix given, not for the one you meant. A determinant returned as 0.0 can be an underflow (two entries of 1e-200 on the diagonal) and is not by itself a proof of singularity: `solve` and `inverse` decide on the exact value. Not for large systems: nothing beyond order 20, no sparse matrices, no least squares, no eigenvalues, no condition number; a 20 × 20 inverse takes about a quarter of a second on a laptop and more when the entries have many different exponents.

- `det(a)` returns the determinant, `solve(a, b)` the vector x with a x = b, `inverse(a)` the inverse, for a matrix given as a list or tuple of rows.
- The elimination is done in exact rational arithmetic on the floats given and each result is rounded once: the conditioning of the matrix costs no digit. A matrix is singular exactly when its exact determinant is zero; no tolerance is involved, so a nearly singular matrix is solved and an exactly singular one is refused.
- Measured on 151 systems (110 well conditioned, 40 badly conditioned, and the Hilbert matrix of order 3 scaled to integers): determinant, solution and inverse equal the exact rational results of SymPy rounded to a float and those of mpmath at 400 digits. NumPy agrees within 1e-12 on the well-conditioned systems; on the badly conditioned ones it loses most digits or refuses the matrix as singular (the numbers are in the evidence file).

- Every function returns floats or raises `ValueError`: a matrix that is not square or larger than 20, entries that are not finite real numbers within 1e100 (booleans refused), a right-hand side of the wrong length, a singular matrix (for `solve` and `inverse`), a result too large for a float.

## 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

## 7. Citation and bill of materials

```
@software{cavazzini_star_linsolve_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-linsolve}: Small linear systems, correctly rounded},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_linsolve/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

## 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts